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# A motion model based on recurrent neural networks for visual object tracking

Mohammad Shahbazi [b,*], Mohammad Hosein Bayat [a], Bahram Tarvirdizadeh [a]

[a] Advanced Service Robots (ASR) Lab, Department of Mechatronics Engineering, Faculty of New Sciences and Technologies, University of Tehran, Tehran, Iran
[b] School of Mechanical Engineering, Iran University of Science and Technology, Tehran, Iran

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ABSTRACT

Object tracking algorithms typically leverage on either or both appearance and motion features of target(s). It is common in multi-object tracking to use both features, whereas the role of motion features in single-object trackers has less been explored. Based on the Long Short-Term Memory (LSTM) architecture of recurrent neural networks, we train a novel motion model to be incorporated into the off-the-shelf single-object trackers. The developed model predicts the target location in each frame based on the history of processed motion features in a few prior frames. This aids the tracking algorithm in dynamically updating the search region location, as apposed to static or probabilistic region settings. We incorporate the model into three state-of-the-art CNN-based trackers, namely GOTURN, SiamFC, and DiMP and illustrate the tracking performance enhancements on popular benchmarks. Significant improvements are achieved specially on the sequences rendering challenging situations such as Low Resolution, Out-of-Plane Rotation, Motion Blur, Fast Motion, and Occlusion. The motion model has a low computational cost and complies with the real-time execution of the base trackers.



## 1. Introduction

Object tracking is among the most important but challenging tasks in computer vision [10,20,21]. With the remarkable progression in the deep learning algorithms specially in the form of convolutional and recurrent neural networks (RNNs), the use of deep features has increasingly been considered in recent object tracking algorithms [17,24]. For a practical and real-time implementation, high accuracy and speed are the major expectations. The main difficulty arises from the fact that trackers are expected to follow a target in a 2D image plane captured from a 3D real scene.

It is common for object tracking algorithms to use either or both appearance and motion features to track a target [2]. Because of the multiplicity of the objects, multi-target tracking algorithms demand more computations [28]. For this class of trackers, the hybrid usage of appearance and motion features is well-established to reduce the search regions' size and increase the appearance model accuracy [24,32]. However, the use of motion features in single-object tracking appears to be considered only optional and has less been explored in the relevant literature [7]. For this class of algorithms, the search region is typically centered at the target's location in the previous frame, albeit with a larger scale to accommodate the target's possible movements. Since the location history (i.e., dynamics) of the target is not considered in the search region settings, the term *static search region* may be used.

The prediction of target location by its motion features brings at least two possibilities: (i) verifying a detected target, and (ii) proposing a more effective search strategy which we refer to it as the *dynamic search region* scheme. Various motion models, both linear and nonlinear, have been studied in the literature mostly for multi-object trackers. A large body of research has focused on the RNN-based models where complex motion features of the targets can be learned by the networks having certain states dealing with the history of movements. Recently, motion models based on the Long Short-Term Memory (LSTM) networks have shown increasingly promising results.

The main motivation behind this study is to explore to what extent the performance of *single*-object trackers can be affected when the dynamic search region scheme through an LSTM-based motion model is considered. We design and train a dedicated LSTM model whose main function is to change the search region settings of the base tracking algorithm (see Fig. 1). Given the motion features of target in the last three frames, the model predicts the target's location in the current frame. This prediction sets the location of search region within which the tracking algorithm could have a higher chance of detecting the target. The model is independent of the target type and motion behavior, since it saves the required motion cues in the embodied intrinsic parameters of the network.

* Corresponding author.
*E-mail address:* shahbazi@iust.ac.ir (M. Shahbazi).

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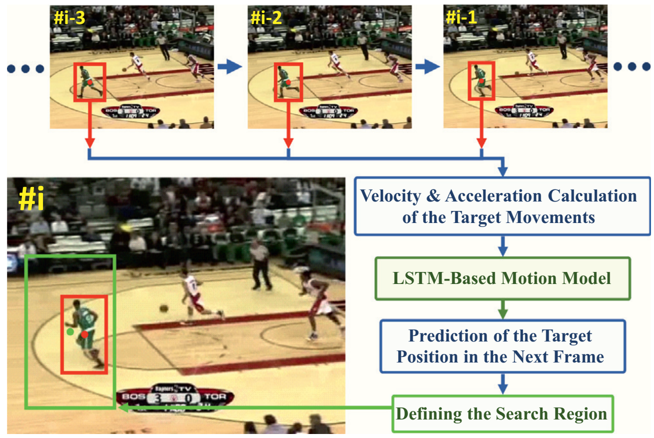



**Fig. 1.** Schematic description of the proposed LSTM-based motion model (*LSTM3*). The trained model uses motion features extracted from the output of object tracking in recent frames to predict the target location in the next frame, based on which the search region location of the tracker is dynamically set.

The rest of this paper is organized as follows. In Section 2, we carefully review related works in the literature. Section 3 presents the architecture and specifications of the developed motion model, and details the training procedure. Evaluation metrics and benchmarks are discussed in Section 4. The influence of the developed model incorporation on the tracking performance is thoroughly studied in Section 5 with respect to the standard and custom metrics. Section 6 presents remarks and an ablation study on the effect of different parameters relevant to the performance and efficacy of the proposed model. Finally, the paper is concluded in Section 7.

## 2. Motion models in existing object tracking algorithms

Motion features are the basis of motion models presented in the literature. These features are typically determined in terms of the bounding box specifications. Associated with each bounding box is a set of primary parameters including: the center location in image plane coordinates $(x, y)$, width and height as the box scale $(w, h)$, as well as its surface area. Moreover, there are secondary/generated features like aspect ratio of the box scale and the rate of each parameter changes, which give more robust and informative features by leveraging historical information from prior frames.

Linear motion models [4,31] are popular in object tracking because they are fast and easy to implement, and they do not need to be trained on enormous data. Kalman filter [13] is a well-known example which is a linear constant velocity model. In [4] a simple yet effective multi pedestrian tracker was proposed using the Kalman filter as the motion model to consider displacements of the targets in subsequent frames. The model provides an estimated location in the next frame for each bounding box considering a number of its features. The object detection framework FrRCNN [23] is used to update the motion model parameters by finding new locations of the targets. The popular Deep Sort algorithm is built on the aforementioned work where the bounding box height and its velocity is additionally included in the motion model [28].

Qiu et al. [22] proposed two motion models exclusively for single-object tracking to change the search region settings. The first one is a simple stochastic *Random Walk* model, and the second model is the *Vector Auto-Regressive*, which needs the real position of the target in some initial frames to train its parameters. Although the second model shows a performance improvement compared to the first one, its reliance on a relatively long sequence of target locations hinders its practical application. Moreover, for a considerable part of tests, no notable improvement has been achieved. We suspect that the reason is on the use of relatively simplistic motion models. In the present study we aim to develop a nonlinear deep learning-based model which is expected to capture higher order dynamics of the target, and its structure is not suffered by naive assumptions such as normal distribution for the transitioning.

It is well-known that in challenging situations like target's complex movements, fast motions, and occlusion the linear models will fail shortly. The relevant literature of recent nonlinear motion models has a clear focus on machine learning. Sadeghian et al. [24] leveraged four LSTMs to track multiple targets by considering the history of triple features: appearance, motion and interaction between objects, each of which is handled by a specific LSTM. The outputs of these networks are stacked at the fourth LSTM to predict the final position of the targets. As for the motion features, only the velocity changes of the targets' positions in adjacent frames from the beginning are considered. Moreover, the function of motion LSTM is to check whether the current frame velocity complies with the “correct” trajectory, meaning that it dose not perform the prediction of target's velocity. The overuse of multiple LSTMs achieves high accuracy in tracking, but prevents the algorithm to be executed in real-time.

Milan et al. [19] proposed an end-to-end model to accomplish multi-target tracking using both recurrent and LSTM neural networks. This method refrains from direct usage of the appearance features and just uses spatial features in tracking. The sole model input is the center position of the targets in image plane. Although this algorithm does not reach the accuracy level of detection-based methods, a high execution speed along with a decent accuracy level makes it appealing for practical applications.

A recent body of research has focused on the extraction of deep appearance features by CNNs, which serve as spatial information instead of bounding box parameters. Bhat et al. [6] proposed a scene-aware architecture that uses spatial features to maximize the target score among background and deceptive objects. A simple RNN is used to preserve temporal information in its states, corresponding to each element in the visual feature map. Wang et al. [27] introduced a motion enhance attention module which calculates the difference between deep visual features for adjacent frames as the weight to obtain better temporal information and motion features. Kashiani and Shokouhi [14] used a motion estimation network to hold the search candidates near the most probable target presence. Although these approaches do exploit spatial information, they do not give stand-alone (independent) dynamic motion models. The spatial information used in these studies is limited to visual features, while other potentially beneficial motion cues like velocity and direction of the target movements are ignored. Moreover, the high calculation cost of extracting deep visual features limits their practicality where high speed processing is demanded.

## 3. LSTM motion model

This section is devoted to the presentation of the proposed motion model. Since the model is based on the LSTM network, the readers are invited to consult the relevant literature (e.g., [11,19]) for a treatment of its concept and architecture. In what follows a detailed illustration of the adopted architecture for motion modeling is presented, and the model training procedure is described.

### 3.1. Model architecture

The developed architecture is termed the LSTM-based *M*otion *M*odel (LSTM3). Given that we have the target position in the last three frames, *detected by the base object tracking algorithm*, the LSTM3 is learned on large data to predict the expected position of the target in the current frame. Based on this prediction, a more precise search region with the most probability of the target presence will be obtained, thus the tracker could demonstrate a better performance in finding and following the target.

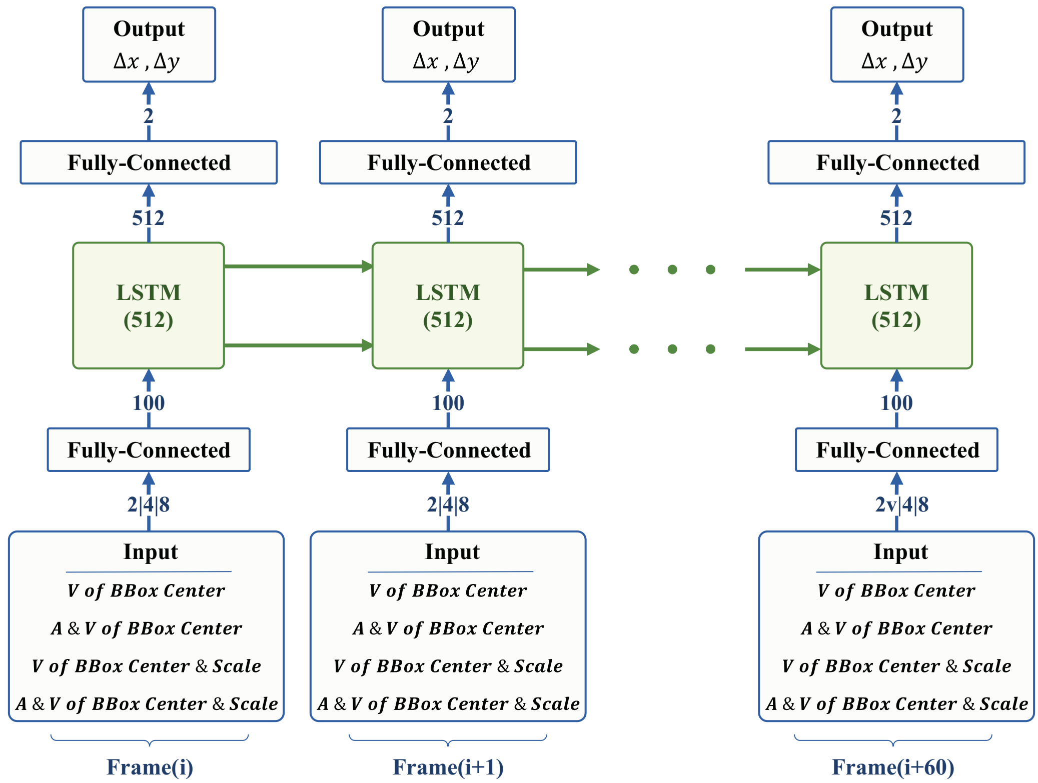



**Fig. 2.** LSTM3 architecture unrolled in the time. Four different input combinations can be formed by the following generated features: acceleration (A) and velocity (V) of bounding box (BBox) parameters, i.e., center location and scale (see also Table 1).

The architecture of LSTM3 is depicted in Fig. 2. The model inputs are different combinations of both velocity and acceleration changes of the bounding box parameters, described in more details in Section 3.1.1. The fully connected and LSTM layers used in the proposed architecture are partially inspired by [1]. A primary fully connected layer with 100 cells prepares the inputs for the LSTM layer with 512 cells. Subsequently, a secondary fully connected layer is used to decrease the dimension of the data. The outputs are $\Delta x$ and $\Delta y$ as the center location changes of the search region in the current processing frame with respect to the target position in the previous frame. Besides, the cell and hidden states of the LSTM, containing the history of the target's movement, are sent to the subsequent frame.

### 3.1.1. Input

The primary motion features considered in our motion model comprise of the bounding box center location in the image plane $P = [x \ \ y]^T$, and the scale feature that contains the width and height of the bounding box $S = [w \quad h]^T$. We use the first and second order changes of these parameters (i.e., the generated features), because this way we get richer and more stable features for network learning and inference. For a prediction in the $i$th frame, define the velocity $V$ (the first-order change), and the acceleration $A$ (the second-order change) of the above-defined features, with the following relations:

$$VP_{i-1} = \begin{bmatrix} x_{i-1} - x_{i-2} \\ y_{i-1} - y_{i-2} \end{bmatrix}, \quad VS_{i-1} = \begin{bmatrix} w_{i-1} - w_{i-2} \\ h_{i-1} - h_{i-2} \end{bmatrix}, \tag{1}$$

$$AP_{i-1} = VP_{i-1} - VP_{i-2}, \quad AS_{i-1} = VS_{i-1} - VS_{i-2}. \tag{2}$$

Therefore, for a prediction in the current frame, the primary features $P$ and $S$ from the last three frames are required. These parameters have already been determined by the base tracking algorithm, and hence, the LSTM3 relies on the outputs of the tracker. In a later section, we show that this reliance could sometimes become problematic, and discuss a technique applied to remedy this issue. Finally, to conduct a comprehensive evaluation, four different combinations of the generated motion features are considered for the training of LSTM3, as summarized in Table 1.

### 3.1.2. Output

The output of the model is the predicted velocity changes of target position in the current frame $VP_i$. It is indeed the transition of bounding box center location from the last to the present frame. Adding this to the target location in the previous frame yields a proposition for the center of search region in the current processing frame. Note that we also analyzed the effect of outputting the bounding box scale changes in addition to the center location changes in our preliminary experiments. This simply yielded no desired results. Perhaps, the reason is that the tested learning-based trackers does not admit this setting because they are trained using search regions with invariant sizes. Nevertheless, we suspect that such an output could improve the tracking performance, if a fresh tracking algorithm is specifically learned to handle this additional feature.

**Table 1**
Different inputs of LSTM3 for comparative evaluation.

| Symbol | Input Features |
|---|---|
| VP | Velocity Changes of Bounding Box Center Location, $VP_{i-1}$ |
| AVP | Acceleration and Velocity Changes of Bounding Box Center Location, $AP_{i-1}$, and $VP_{i-1}$ |
| VPS | Velocity Changes of Bounding Box Center Location and Scale, $VP_{i-1}$, and $VS_{i-1}$ |
| AVPS | Acceleration and Velocity Changes of Bounding Box Center Location and Scale, $AP_{i-1}$, $VP_{i-1}$, $AS_{i-1}$, and $VS_{i-1}$ |

### 3.2. Training procedure

The LST*M3* is trained using the combination of two popular training datasets: ALOV300+ [25] which is specific for single-object trackers, and MOT16 [18] as a training set for multi-object tracking. ALOV300+ has 315 sequences labeled every five frames, 89,364 labels in total. We use linear interpolation with random choices to annotate intermediate frames. MOT16 has 7 sequences with a total of 199,960 labels for most of the targets in each frame. We treat this collection as if it is a single-object training set by separating all labels related to each target. Data augmentation is applied to the concatenated datasets by reversing all the sequences. In the next step, the motion features of the labels are obtained following the procedure explained in Section 3.1.1 and using (1) and (2). With wrapping the labels in groups of the same size, we finally got 511,740 training labels in total.

The whole training procedure is done offline. However, it is needed for the inference of model that the particular parameters of the LSTM preserve the history of target motion cues and share it with the future frames. As such, the model updates online and adapts with the target movements changes. Moreover, it is common in RNN-based architectures to reinitialize their parameters after an optimal time step, to avoid error accumulation caused by false predictions. Accordingly, we reset the LSTM parameters, cell and hidden states, after every 60 frames, the value determined in the empirical analysis for the best LST*M3* performance.

## 4. Evaluation procedure and metrics

The performance of the developed motion model will be illustrated on three state-of-the-art single-object trackers. We first describe the specifications of evaluation metrics and protocols pertinent to considered benchmarks and motivate some custom-defined metrics. Then, the selected object trackers are shortly introduced, and the procedure for incorporating the motion model is discussed.

### 4.1. Benchmarks

For the empirical evaluations, two versions of the commonly referenced Online Object Tracking Benchmark (OTB), namely OTB50 [29] and OTB100 [30], as well as the Visual Object Tracking Benchmark (VOT) [15], are considered. OTB50 and OTB100 contain 50 and 100 challenging sequences, respectively. Each sequence is annotated frame-by-frame and 11 challenge attributes, described later in Section 5.1.2, are designated. In these benchmarks, two metrics are introduced for evaluation: the Center Location Error (CLE) which is a measure for the precision of tracking, and the Bounding Box Overlap, parameterized by the intersection over union (IOU) which determines the success rate of tracking.

Following the protocol presented in [29,30], precision and success curves can be plotted using the CLE and IOU metrics, respectively. The precision plot of each sequence reports the percentage of frames for which the CLE magnitude is less than a certain threshold. Spanning the threshold value typically from 0 to 50 pixels, the precision curve is obtained for each test sequence, and the average of all sequences is used for the overall precision illustration. Similarly, the success curve is plotted by determining the percentage of frames whose IOUs are greater than a threshold being in the range of 0 to 1 by definition.

The third evaluation benchmark, VOT, including 60 sequences, is also considered because it includes diverse challenges that are not similar to the training data and the OTB benchmarks. In general, VOT can use *reset-based* and *no-reset-based* approaches to evaluate the tracking performance. Similar to OTB, in the no-reset-based approach the tracker does not re-initialize during the tracking, even with possibly wrong predictions and failures. In contrast, the reset-based approach re-initializes the tracker after five frames of each tracking failure. Tracking performance is measured by three indicators: accuracy, robustness and their combination, all based on IOU metric. Accuracy is the overall average overlap, and robustness is measured by counting the number of failures as zero overlap. Their combination obtained by the Expected Average Overlap or EAO for short (please refer to [15] for more details).

As illustrated in Section 3, the significance of the proposed motion model LST*M3* lies on its use of the motion history features in tracking. Since the VOT toolkit reports follow the reset-based scheme, every time it resets the tracker the LST*M3* is forced to forget the valuable information of target behaviors developed in its parameters. As such, the LST*M3* might perform poorly in helping the tracker to re-detect the target after failures using a dynamic search region. Therefore, one can see that the default VOT toolkit reports are not sufficiently illustrative of the performance enhancement by the LST*M3* addition. To compensate for this shortcoming, the performance of the motion model is additionally evaluated following a no-reset-based scheme.

Three factors are considered in the no-reset-based scheme to investigate the tracking accuracy and robustness: Average Overlap (AO), zero IOU percentage, and target missing-recovery. The target missing events occur in frames whose IOUs are lower than a certain threshold defining the margin of tracking failure. Accordingly, if the IOU restarts to grow and becomes larger than the threshold, the target is assumed to be recovered, and the average number of frames between each two missing and recovery ($\Delta$*Frames*) in each sequence is recorded. Robustness curve can then be plotted using IOU thresholds from 0 to 20 percent and overall average of $\Delta$*Frames* for all sequences, in which the Area Under Curve (AUC) can summarize the tracking robustness.

### 4.2. Selected trackers

The relevance and efficacy of the proposed motion model can best be demonstrated when incorporated into tracking algorithms. To this end, we opted three state-of-the-art single-object trackers, whose main characteristics are described in the following. Note that these particular trackers are selected because they are well-known for their real-time compatibility among state-of-the-art deep trackers.

The Generic Object Tracking Using Regression Networks (**GOTURN**) [9] is a well-known deep learning-based tracker, built on CaffeNet [12], with a reasonable accuracy and real-time compatibility. The network input is the crops from two adjacent frames. The first crop is the detected bounding box in the first frame padded with twice of the box dimensions to account for the surrounding environment and possible movements of the target. The spatial properties of second crop is the same as the first, but captured from the current frame; therefore, the algorithm originally follows a basic static search region scheme.

**SiamFC** [3], developed based on fully convolutional Siamese networks, is another well-known object tracker to be tested in this study. It is the winner of VOT17 in the task of real-time challenge [16]. SiamFC receives a template in the first frame as its input and locates it in a larger segment in the subsequent frames. Here again, the search region location in each frame is solely determined by the location of the detected template in the previous frame.

The Discriminative Model Prediction (**DiMP**) [5], as a learning-based generic object tracking method, is the last tracker used for evaluation. In contrast to generative-based tracking approaches such as SiamFC, DiMP focuses on updating the target template and utilizing the object information by discriminating it from the background through a classification. Its architecture utilizes two branches: classification and accurate bounding box prediction [8]. Like previous trackers, this architecture uses static search region in current frame.

For these trackers, we use LST*M3* to modify the strategy of searching by *dynamically* locating the search region based on the historical features of the target movements. More specifically, the motion model receives the spatial features of the target detected by the base tracker in the last three frames and predicts its location in the current frame. The location of the search region for the tracker is updated accordingly. As will be shown in the next section, the performance of considered trackers is enhanced when implementing this dynamic search region scheme, thanks to the accurate predictions provided by the trained LST*M3*.

## 5. Evaluation results

In this section the results of all trackers evaluation on the considered benchmarks are illustrated. The first two sections are devoted to the evaluations on OTBs and VOT benchmarks following the default protocols and toolkit. This is followed by an additional evaluation for all collections based on the custom metrics motivated in Section 4.1.

### *5.1. Evaluation on OTB50 and OTB100*

Below the overall performance of trackers as well as their performance on challenges are discussed on OTB50 and OTB100 benchmarks.

#### *5.1.1. Overall performance*

Using the test sequences from OTB50 and OTB100, the precision and success plots for the original and upgraded GOTURN trackers are obtained and depicted in Fig. 3. The results are obtained with the LST*M3* of type AVPS (see Table 1), the one exhibited superior performance in the analysis which will be discussed in Section 6. As can be seen, precision and success rate of the original tracker are elevated respectively by 9% and 6.2% for OTB50 and by 8% and 6.8% for OTB100. Note that the reported values in the legend of precision plots are the scores corresponding to the threshold equal to 20 pixels. Also, AUC is calculated for the success curves and reported in the legend of respective panel. Finally, the term “Clip” in the legends refers to an input clipping technique addressed in Section 6.2.

Similar evaluations are conducted for the SiamFC and DiMP trackers equipped with the best type of LST*M3* as mentioned earlier. Fig. 4 illustrates the precision and success plots of SiamFC evaluation on OTBs. Here again, the performance of upgraded SiamFC has significant improvements compared to its original version. Whereby, respectively for OTB50 and OTB100, 3.9% and 3.6% progression are achieved for precision as well as 1.9% and 1.6% for tracking success, demonstrating the generality of the proposed motion model. Nevertheless, for high magnitudes of overlap threshold in Fig. 4b and Fig. 4d, no remarkable improvement in the success rate is observed. This indicates that the LST*M3* may not be considerably effective on the SiamFC performance for the high precision interval. Fig. 5 shows the evaluation results on DiMP tracker. Like previous trackers, there is significant growth in its

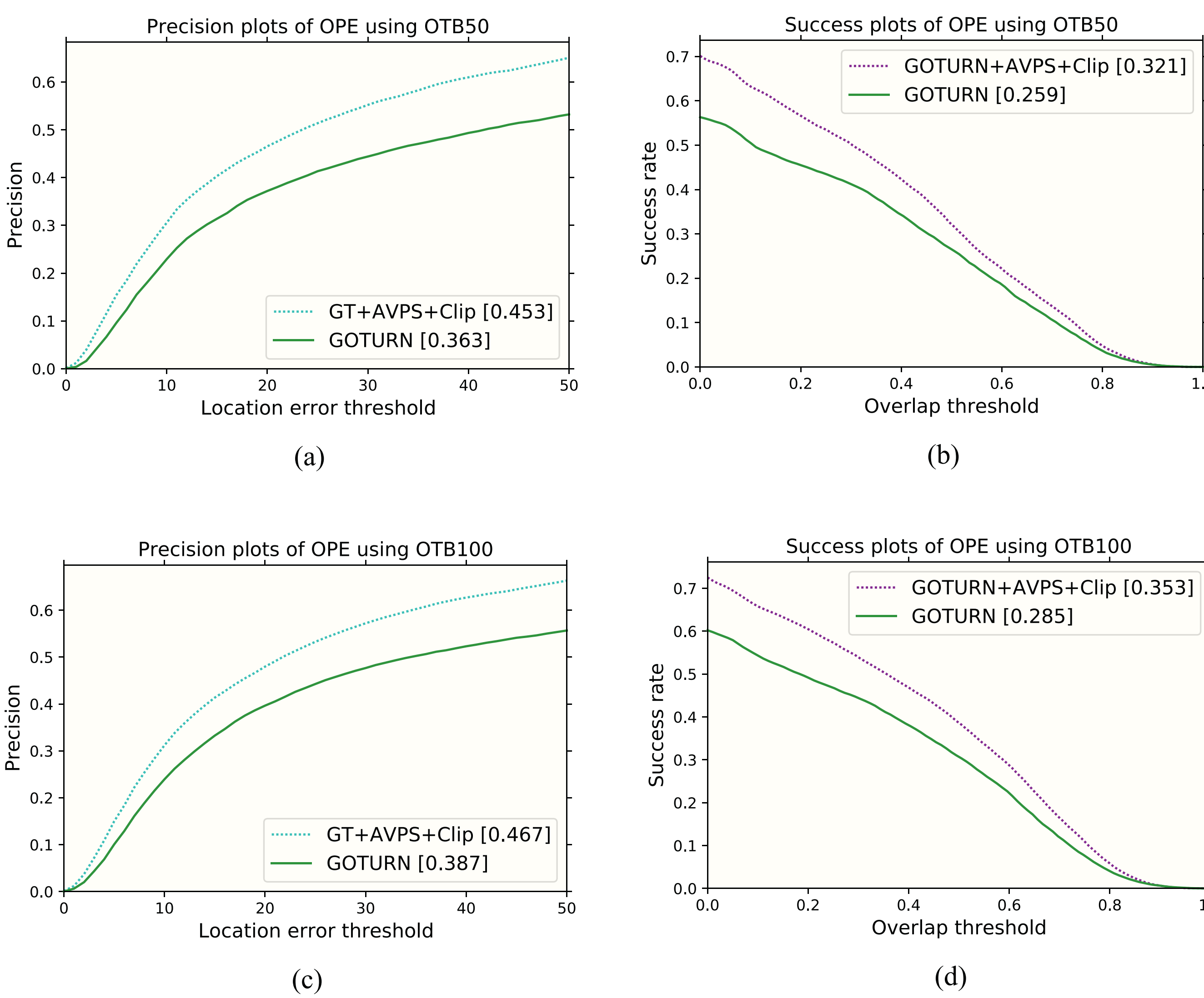



**Fig. 3.** (a,c) Precision and (b,d) success plots of original and upgraded GOTURN using the motion model with the best performance.

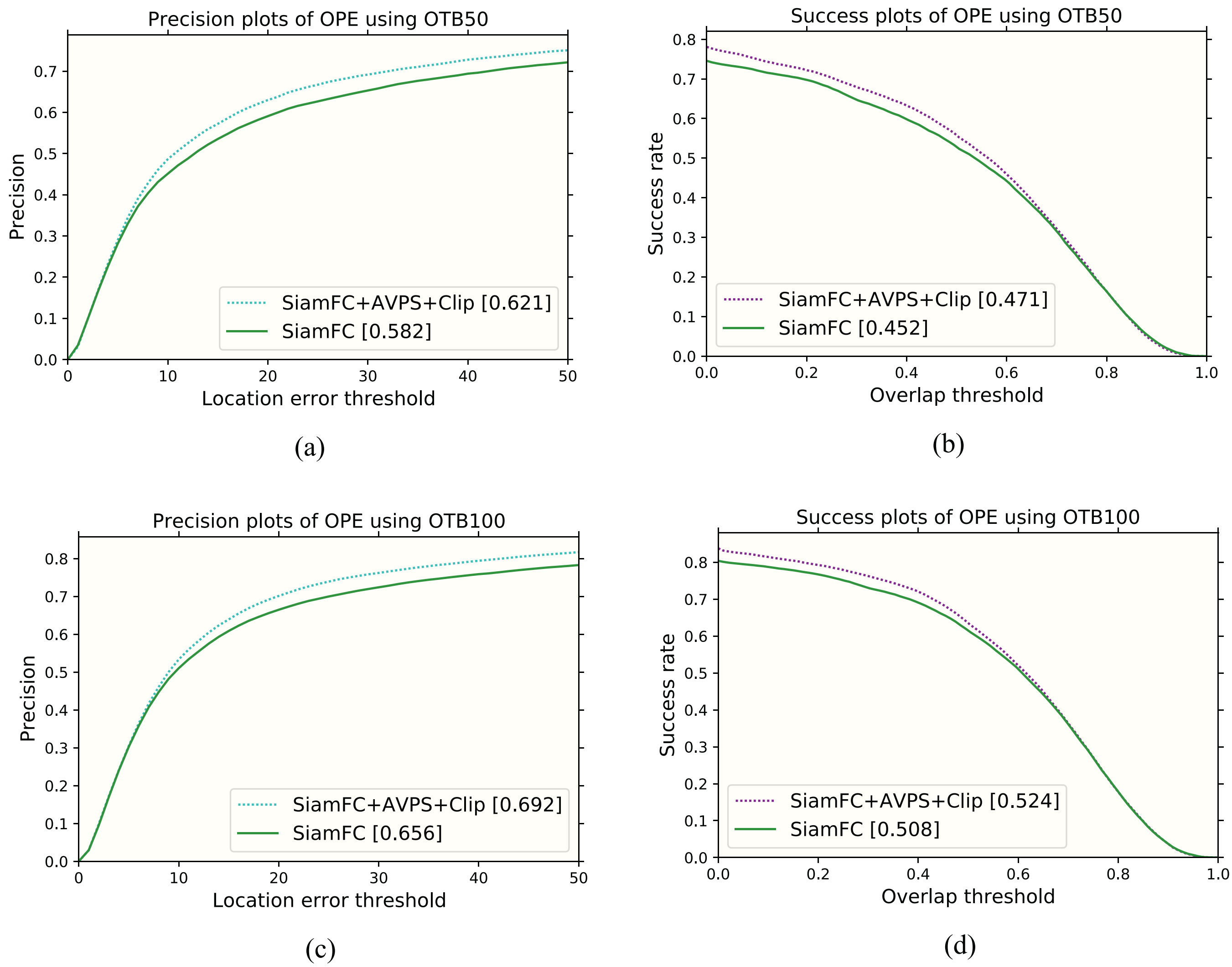



**Fig. 4.** (a,c) Precision and (b,d) success plots of original and upgraded SiamFC using the motion model best performed on GOTURN evaluation.

performance by motion model addition. Finally, Table 2 summarizes progression values for all trackers.

*5.1.2. Evaluation on OTB challenges*

It is of a particular interest to study the influence of the proposed motion model on facing the specific challenges categorized in the OTB benchmark and reported here in Table 3. For each attribute, we compare the evaluation metrics of the original trackers and the ones modified by the best type of LST*M3*, namely AVPS. Fig. 6 summarizes the whole results, upper panels for the GOTURN, middle panels for SiamFC and the lower panels for the DiMP.

As can be seen, the LST*M3* contribution enhances the performance in dealing with almost all the challenge attributes for all trackers. The best result with the GOTURN is obtained in the *Low Resolution (LR)* challenge by 19% improvement in precision and 13% in success scores. As for the SiamFC, one can observe that the highest rise in precision is 7% and 4% improvement in the success score for the *Motion Blur (MB)* challenge. Of particular significance is the improvement achieved for the *Fast Motion (FM)* challenge, which indicates that the LST*M3* is well trained to accurately predict even fast and drastic motions. For the SiamFC, although there are a few challenges for which the contribution of the proposed motion model is not substantial, none of the challenge attributes is negatively affected by this modification. This implies that the incorporation of the LST*M3* into this tracker is, at the worst scenario, just safe. Like other trackers, significant improvements are achieved for DiMP specially in *Deformation (DEF)* and *Low Resolution (LR)* challenges, up to 10% enhancement for precision and 7% for success.

Notice the difference in the amount of enhancement offered by the LST*M3* when comparing the trackers on similar challenge attributes. This strongly relates to the capability of the tracking algorithm itself in handling that challenge. In this study, we limited the modification of the base tracking algorithm to the search region location setting. A full incorporation of the LST*M3* could possibly change the way a specific challenge is addressed by the base algorithm by jointly considering the appearance and motion features.

*5.2. Evaluation on VOT2018*

The VOT default toolkit (reset-based) evaluation is used for performance illustration. The first three rows in Table 4 show the accuracy, robustness and their combination accordingly. As discussed earlier, a no reset-based evaluation is additionally performed whose main results are summarized in the last row of the same table. As can be seen, when this scheme is followed, the motion model better contributes in the performance by leveraging on the motion history endowed in its learned parameters.

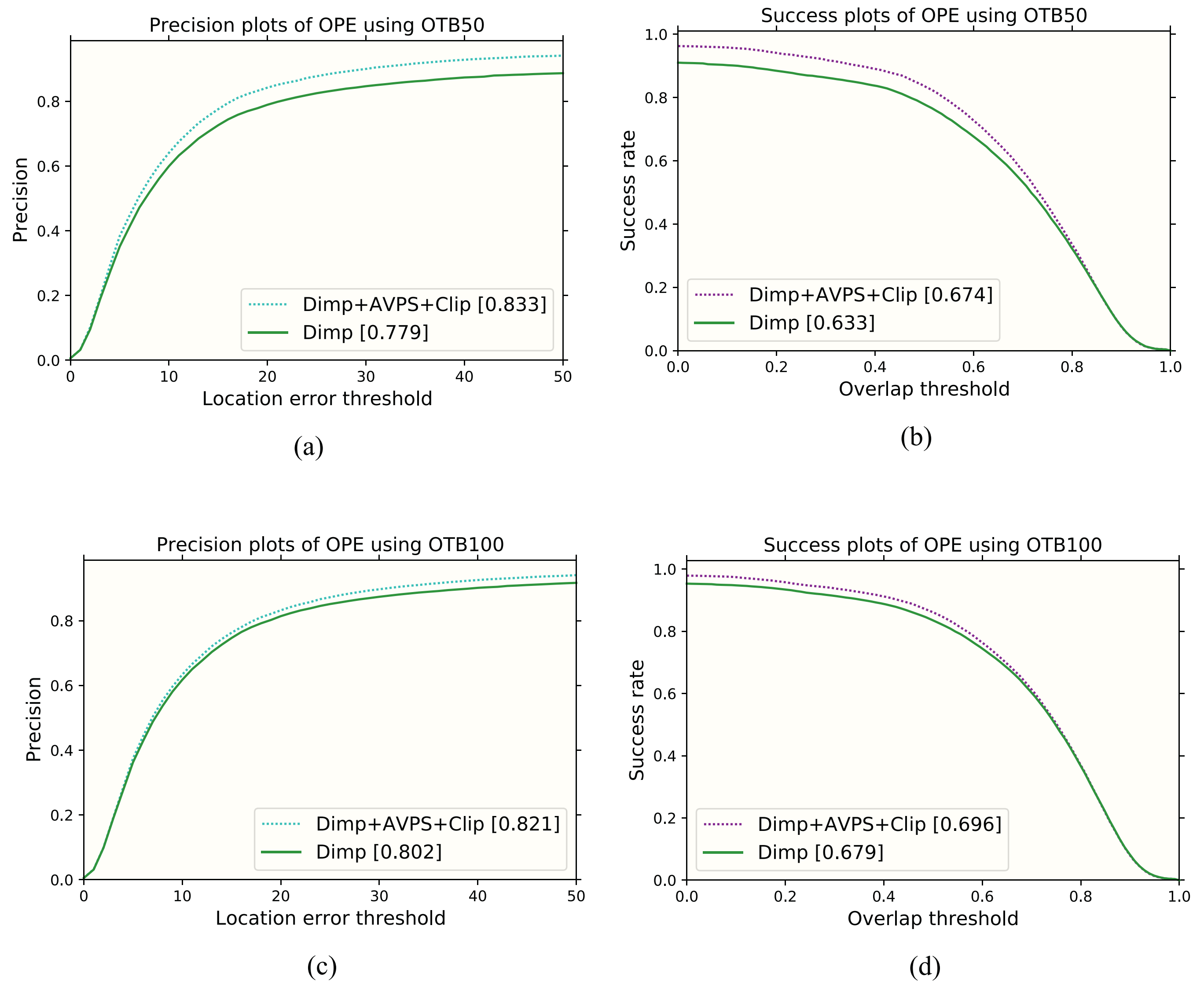



**Fig. 5.** (a,c) Precision and (b,d) success plots of original and upgraded DiMP using the motion model best performed on GOTURN evaluation.

### *5.3. Robustness evaluation using custom-defined metrics*

A more detailed evaluation of the no reset-based scheme is presented here for all the three benchmarks and using the custom metrics defined in Section 4.1.

The results of overall Average Overlap (AO), reported in Table 5 for all benchmarks, show remarkable progression in tracking performance by using LST*M3*. Additionally, the percentages of zero overlap, summarized in Table 6, confirm robustness enhancement using the LST*M3*. Here, there is an exception for SiamFC algorithm on VOT collection; however, for all the other cases considerable improvements are

**Table 2**
Percentage of enhancements for precision and success scores by LST*M3*, using OTB50 and OTB100.

| Benchmark | Enhancement | GOTURN | SiamFC | DiMP |
|---|---|---|---|---|
| OTB50 | Precision | 9 | 3.9 | 5.4 |
| | Success | 6.2 | 1.9 | 4.1 |
| OTB100 | Precision | 8 | 3.6 | 1.9 |
| | Success | 6.8 | 1.6 | 1.7 |

**Table 3**
The description of challenge attributes, present in the OTB sequences. We evaluate our motion model performance in solving these challenges using precision and success metrics.

| Attribute | Explanation |
|---|---|
| IV | Illumination Variation: the illumination in the target region is significantly changed. |
| SV | Scale Variation: the ratio of the bounding boxes of the first frame and the current frame is out of the range $[1/t_s, t_s]$, $t_s > 1$. |
| OCC | Occlusion: the target is partially or fully occluded. |
| DEF | Deformation: non-rigid object deformation. |
| MB | Motion Blur: the target region is blurred due to the motion of target or camera. |
| FM | Fast Motion: the motion of the ground truth is larger than $tm$ pixels ($tm = 20$). |
| IPR | In-Plane Rotation: the target rotates in the image plane. |
| OPR | Out-of-Plane Rotation: the target rotates out of the image plane. |
| OV | Out-of-View: some portion of the target leaves the view. |
| BC | Background Clutters: the background near the target has the similar color or texture as the target. |
| LR | Low Resolution: the number of pixels inside the ground-truth bounding box is less than $tr$ ($tr = 400$). |

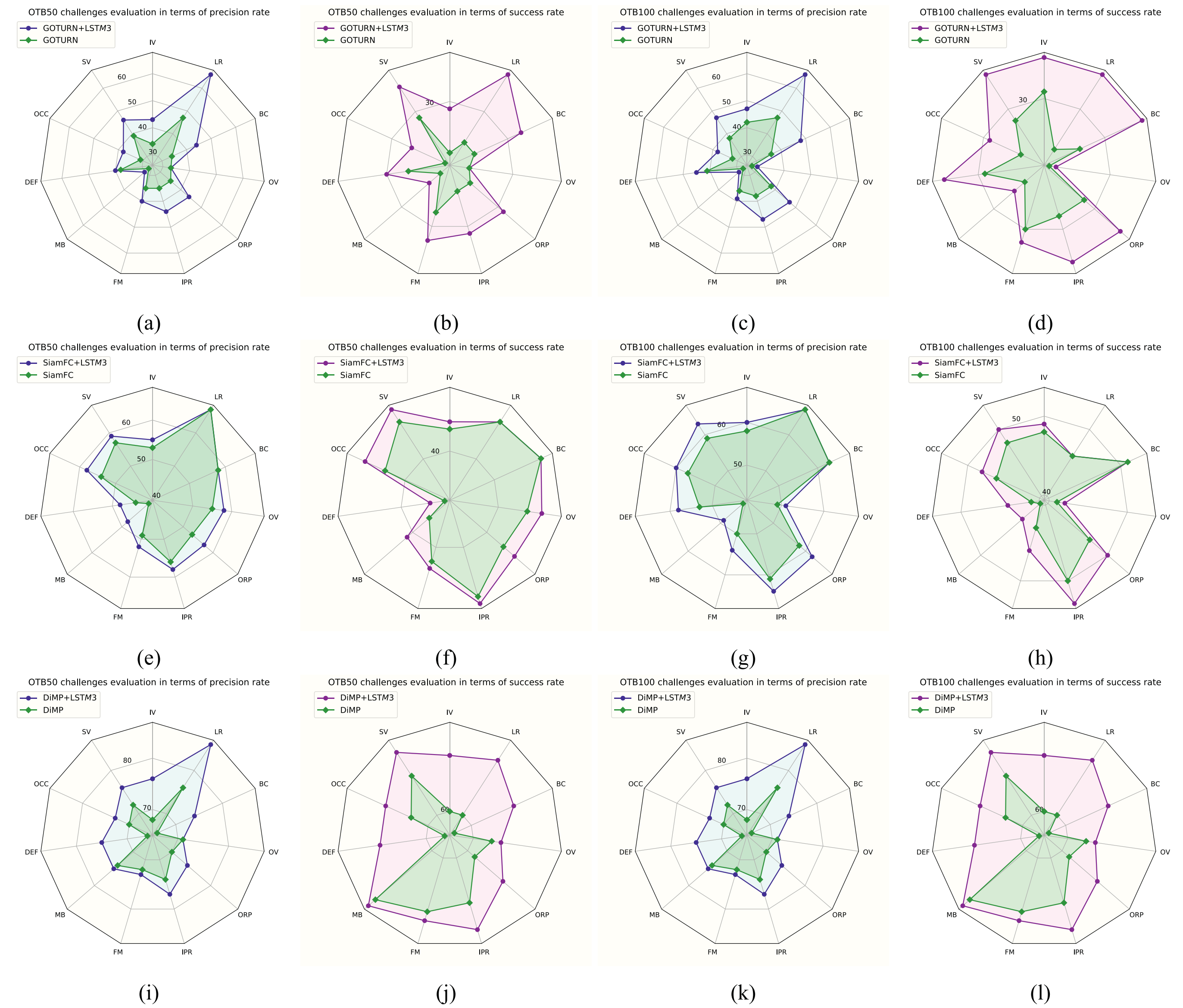



**Fig. 6.** Evaluation of OTB50 and OTB100 challenges for GOTURN, SiamFC, and DiMP object tracking algorithms, using LS*TM3* with the best performance, i.e., model with the full features input.

observed. Finally, Fig. 7 depicts robustness curves based on the average Δ*Frames*, i.e., the average number of frame between each target missing-recovery (see Section 4.1), when the IOU threshold is spanned from 0 to 20. AUC is reported as robustness score for each tracker with and without motion model. Also the average of Δ*Frames* for zero IOU as well as all thresholds are detailed in Table 7. Overall, these results also support that the addition of LS*TM3* to the base trackers improves the performance in most cases.

**Table 4**
The VOT2018 challenge results. The Average Overlap (AO) is used as the metric of no reset-based scheme.

| Tracker | GOTURN | GOTURN + LSTM3 | SiamFC | SiamFC + LSTM3 | DiMP | DiMP + LSTM3 |
|---|---|---|---|---|---|---|
| Accuracy (↑) | 0.407 | 0.425 | 0.472 | 0.479 | 0.570 | 0.571 |
| Robustness (↓) | 3.955 | 4.366 | 2.028 | 2.143 | 0.940 | 0.964 |
| EAO (↑) | 0.136 | 0.119 | 0.172 | 0.181 | 0.370 | 0.375 |
| Unsupervised AO (↑) | 0.195 | 0.233 | 0.348 | 0.339 | 0.444 | 0.454 |

## 6. Ablation study and remarks

In what follows we discuss a number of parameters influencing the effectiveness of the proposed motion model by using GOTURN tracker as the baseline and OTB50 benchmark.

### 6.1. Evaluation of motion model inputs

As explained in Section 3.1.1, the LS*TM3* is separately trained by four different combinations of input features. Accordingly, the testings are performed individually on all of them for a comparative evaluation as

**Table 5**
Robustness analysis on OTB50, OTB100 and VOT2018 using Average Overlap (AO).

| Benchmark | GOTURN | GOTURN + LSTM3 | SiamFC | SiamFC + LSTM3 | DiMP | DiMP + LSTM3 |
|---|---|---|---|---|---|---|
| OTB50 | 25.92 | 32.08 | 45.18 | 47.07 | 63.17 | 67.24 |
| OTB100 | 28.52 | 35.32 | 50.77 | 52.41 | 67.80 | 69.48 |
| VOT2018 | 25.36 | 25.71 | 33.51 | 33.12 | 54.41 | 56.64 |

**Table 6**
Robustness analysis on OTB50, OTB100 and VOT2018 using percentage of zero IOUs.

| Benchmark | GOTURN | GOTURN + LSTM3 | SiamFC | SiamFC + LSTM3 | DiMP | DiMP + LSTM3 |
|---|---|---|---|---|---|---|
| OTB50 | 43.68 | 29.88 | 25.42 | 21.10 | 9.02 | 3.75 |
| OTB100 | 39.78 | 27.47 | 19.61 | 16.21 | 6.70 | 2.12 |
| VOT2018 | 47.412 | 47.415 | 35.7 | 36.97 | 15.78 | 11.21 |

reported in Fig. 8. Overall, the tracking performance significantly enhances by the addition of motion model, while the difference between the different implementations of LST*M3* is subtle. For both metrics, the motion model with the full input, symbolized by AVPS in Table 1, shows the best results. In this case, the precision and success rate of the original tracker are elevated by 6.4% and 5.3%, respectively. Note that the success rates are determined by the AUC metric. Even for the motion model with a minimal input combination VP as defined in Table 1, one can see notable improvements in both metrics corroborating the effectiveness of the LST*M3*.

### 6.2. Thresholding in the model input

The detected bounding box by the object tracker in each frame is used to calculate the velocity and acceleration changes of the target movements, which are served as the input of LST*M3*. Any false or inaccurate detection by the tracking algorithm, likely to occur due to various common disturbing situations, will simply result in a misleading prediction by the developed motion model. Consequently, a wrong search region will be proposed to the tracker causing further tracking failures.

To remedy this issue, we apply a simple clipping technique on the LST*M3* input using the moving average of the motion features. In doing so, we determine the overall average of each motion feature over the past frames. Then in the present frame, every generated feature based on the tracker output is compared with its overall average. In case of exceeding a predefined threshold, the intended feature value will be clipped by one and a half of the corresponding average value. Somewhat surprisingly, this simple clipping technique considerably improves the tracking performance, as demonstrated by the precision and success plots in Fig. 9.

A more detailed illustration is given in Fig. 10 for a sequence of OTB in which the tracking behavior of upgraded GOTURN is investigated when the input clipping is applied. In this figure the center location of three boxes are plotted against the frame number, namely the target ground-truth, the detected bounding box by the tracker, and the search region which is predicted by the LST*M3*. As seen in Fig. 10a, when no clipping is applied the target is lost by the tracker at a very early stage (approximately in frame #20). Consequently, the LST*M3* is provided by false inputs leading to conflicting search region propositions that

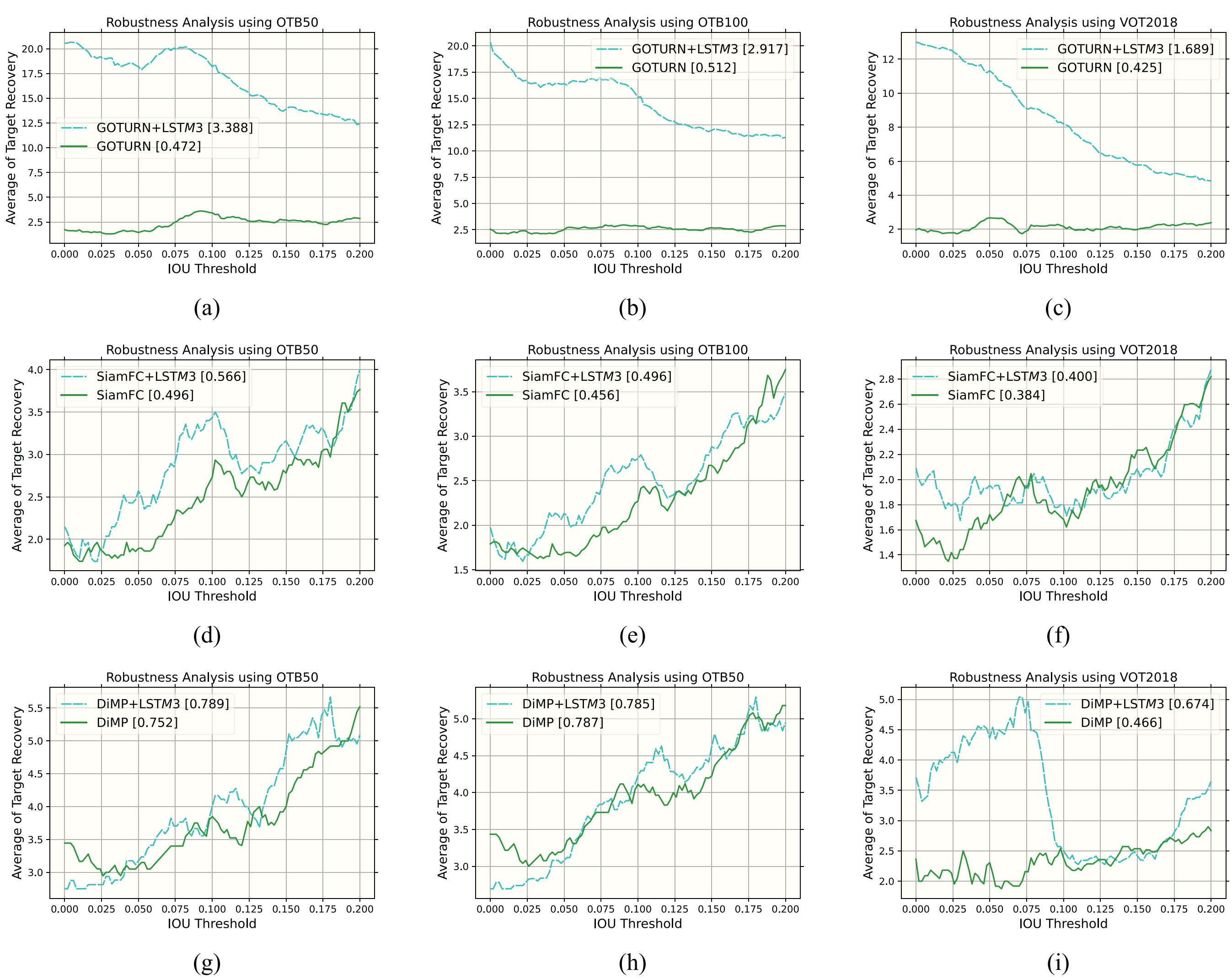
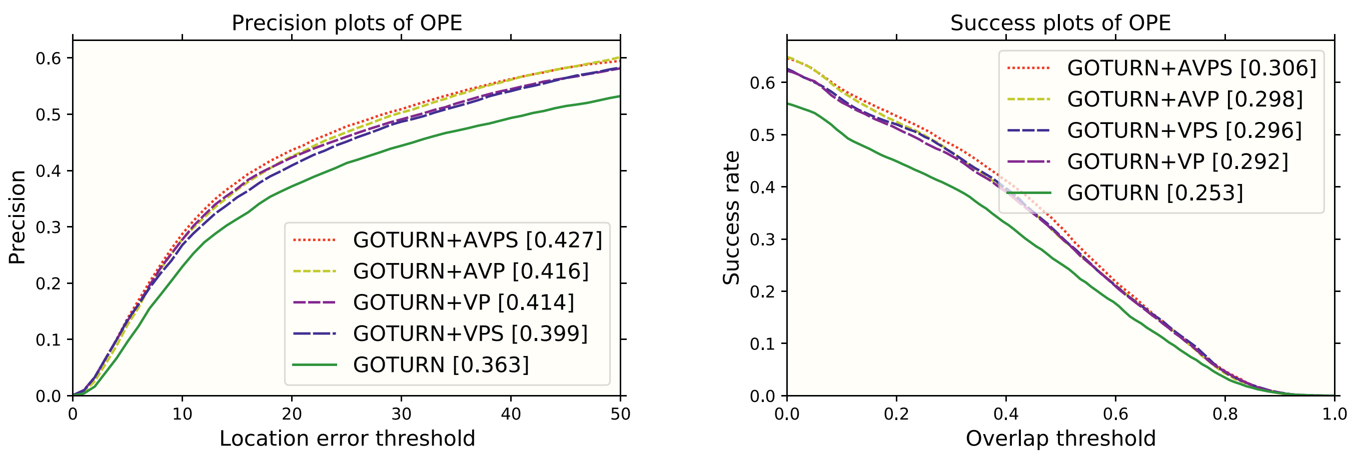



**Fig. 7.** Robustness analysis of GOTURN, SiamFC and DiMP object tracking algorithms, using LST*M3* with the best performance, i.e., model with the full features input, on OTB50, OTB100 and VOT2018. The average of target missing/recovery, symbolized as $\Delta Frames$, is defined in Section 4.1.

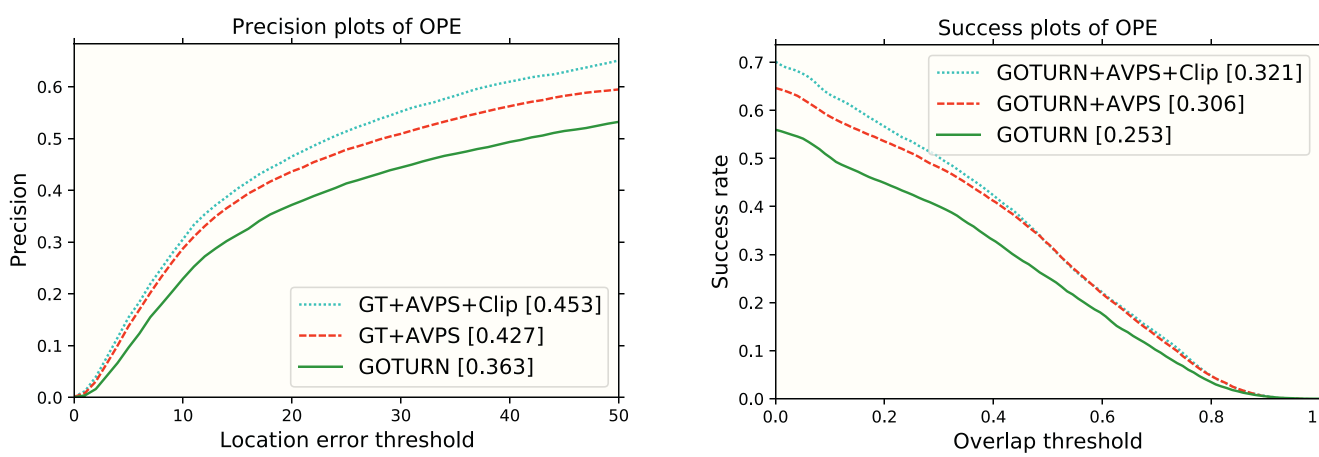
**Table 7**
Robustness analysis on OTB50, OTB100 and VOT2018 using the average Δ*Frames*. Please refer to Section 4.1 for the definition.

| Benchmark | Threshold | GOTURN | GOTURN + *LSTM3* | SiamFC | SiamFC + *LSTM3* | DiMP | DiMP + *LSTM3* |
|---|---|---|---|---|---|---|---|
| OTB50 | IOU = 0 | 23.92 | 3.62 | 34.17 | 21.62 | 31.32 | 22.18 |
| | Average of IOU = [0,0.2] | 12.58 | 4.82 | 22.25 | 12.73 | 27.4 | 14.86 |
| OTB100 | IOU = 0 | 23.76 | 7.76 | 42.91 | 36.48 | 25.43 | 16.92 |
| | Average of IOU = [0,0.2] | 16.27 | 9.81 | 31.01 | 22.98 | 2.53 | 11.50 |
| VOT2018 | IOU = 0 | 26.84 | 6.23 | 29.87 | 33.71 | 29.77 | 32.56 |
| | Average of IOU = [0,0.2] | 17.80 | 11.06 | 23.38 | 29.97 | 22.80 | 21.25 |




(a) (b)

**Fig. 8.** (a) Precision and (b) success plots of overall performance by incorporating *LSTM3* with different inputs to GOTURN object tracker, evaluated on OTB50 using One Pass Evaluation (OPE) protocol explained in [29].



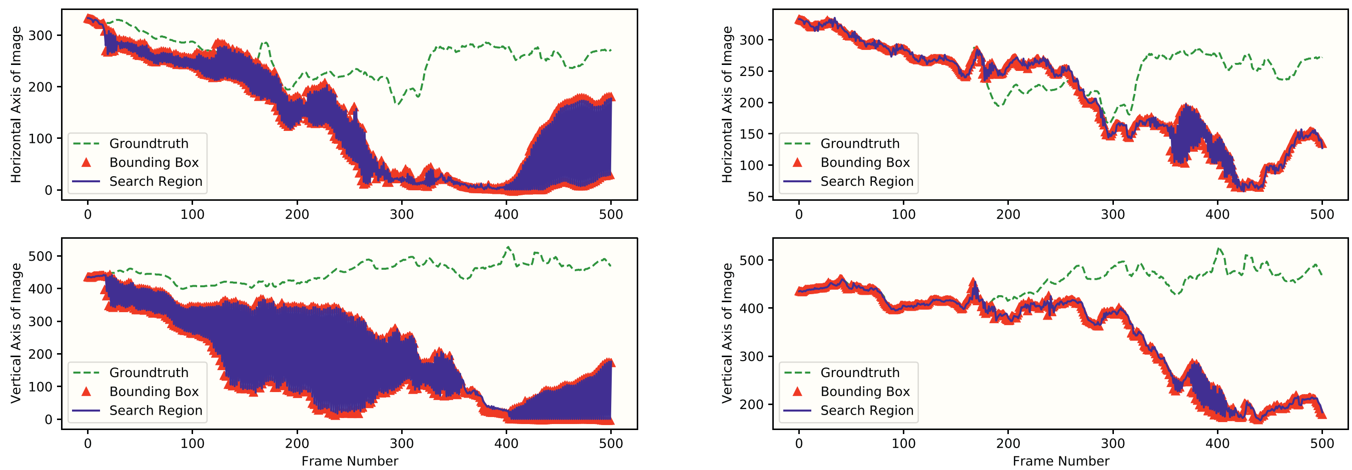

(a) (b)

**Fig. 9.** Clipping the input of *LSTM3* to attenuate the effect of tracker false detections on the motion model performance.

cause drastic fluctuations. When the clipping is applied in Fig. 10b, the amplitude of these bounces is significantly decreased, and the tracker could detect the target more precisely and for a longer sequence. A stricter clipping could further eliminate the fluctuations, however, at the risk of losing the target in fast and drastic motions.

### *6.3. Processing speed analysis*

It is important to study the influence of *LSTM3* addition on execution speed of each tracker. The speed of trackers in terms of Frame Per Second (FPS) for the original and upgraded tracking algorithms with *LSTM3* are reported in Table 8. All tests were run on the Intel Core i7-8700X CPU @ 3.7GHz and an Nvidia GeForce GTX 1030. The results imply that the motion model addition does not affect the processing speed considerably, since the whole training process of the motion model is done offline. The average of run-time for the motion model was 28 ms for all tests. Note that the current implementation runs the motion model on the CPU, without any optimization and parallelization like TensorRT or GPU accelerations. Such optimizations could further improve the processing speed.




(a) LST*M3* without input clipping (b) LST*M3* with input clipping

**Fig. 10.** An example of applying clipping to LST*M3* input, tested on an OTB sequence.

**Table 8**
Speed analysis for all trackers with and without LST*M3*. G, S, D, and M refer to GOTURN, SiamFC, DiMP, and Motion Model, respectively.

| Tracker | G | G + M | S | S + M | D | D + M |
|---|---|---|---|---|---|---|
| FPS | 28.12 | 26.13 | 9.7 | 9.85 | 18.98 | 19.8 |

## 7. Conclusion

A nonlinear LSTM-based motion model for visual object tracking has been presented in this paper. The model is trained on a combination of large annotated video collections to predict the target position in each frame, given the target motion features in a few prior frames. We used this model to modify the search region settings in three state-of-the-art single object trackers. Empirical evaluations of the modified tracking algorithms on the benchmarks OTB50, OTB100, and VOT demonstrate a significant enhancement in the tracking performance thanks to the accurate search region propositions by the developed motion model. The upgraded trackers by this motion model outperform the original trackers in handling almost all tracking challenges, most notably in Low Resolution, Out-of-Plane Rotation, Motion Blur, Fast Motion, and Occlusion. The presented real-time compatible model is trained end-to-end, requires no online fine-tuning, and updates motion cues of arbitrary objects in its intrinsic parameters.

This study provides concrete evidence for the relevance of motion models in the paradigm of single object tracking, the problem that was less explored in previous works. The concurrent training of the tracker and motion model, and the customization of tracking algorithms to better accommodate for the motion model capabilities are among the future research directions.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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